Customer Segmentation using K-Means Clustering

End-to-end data analytics workflow to identify customer groups and enable data-driven business decisions

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Project Objective

Identify Groups

Discover distinct customer segments based on behavior

Pattern Analysis

Understand spending patterns across demographics

Business Decisions

Enable data-driven strategies for customer engagement

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Dataset & Features

Source

- Real dataset (customers.csv)
- Synthetic fallback with make_blobs

Features

- Age
- Spending Score (1–15,000)

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Tools & Technologies

Python

Core programming language

Pandas & NumPy

Data manipulation and analysis

Matplotlib & Seaborn

Visualization libraries

Scikit-learn

K-Means clustering implementation

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Project Workflow

01

Data Loading

CSV import or synthetic generation

02

Visualization

Scatter plot analysis

03

Feature Selection

Age and spending score

04

Elbow Method

Optimal cluster determination

05

Clustering

Model training with K=4

06

Visualization

Color-coded segments

07

Labeling

Business-friendly categories

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Elbow Method Analysis

Key Insight

Elbow point observed at **K = 4**

Optimal number of clusters identified through inertia calculation

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Customer Segmentation Results

High Spenders (Young)

50 customers

Moderate Spenders

60 customers

Low Spenders (Older)

45 customers

High Spenders (Older)

45 customers

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Key Insights & Interpretation

4 Distinct Groups

Clear separation based on age and spending behavior

High-Value Detection

Identified valuable customer segments for targeting

Visual Patterns

Patterns clearly visualized using clustering techniques

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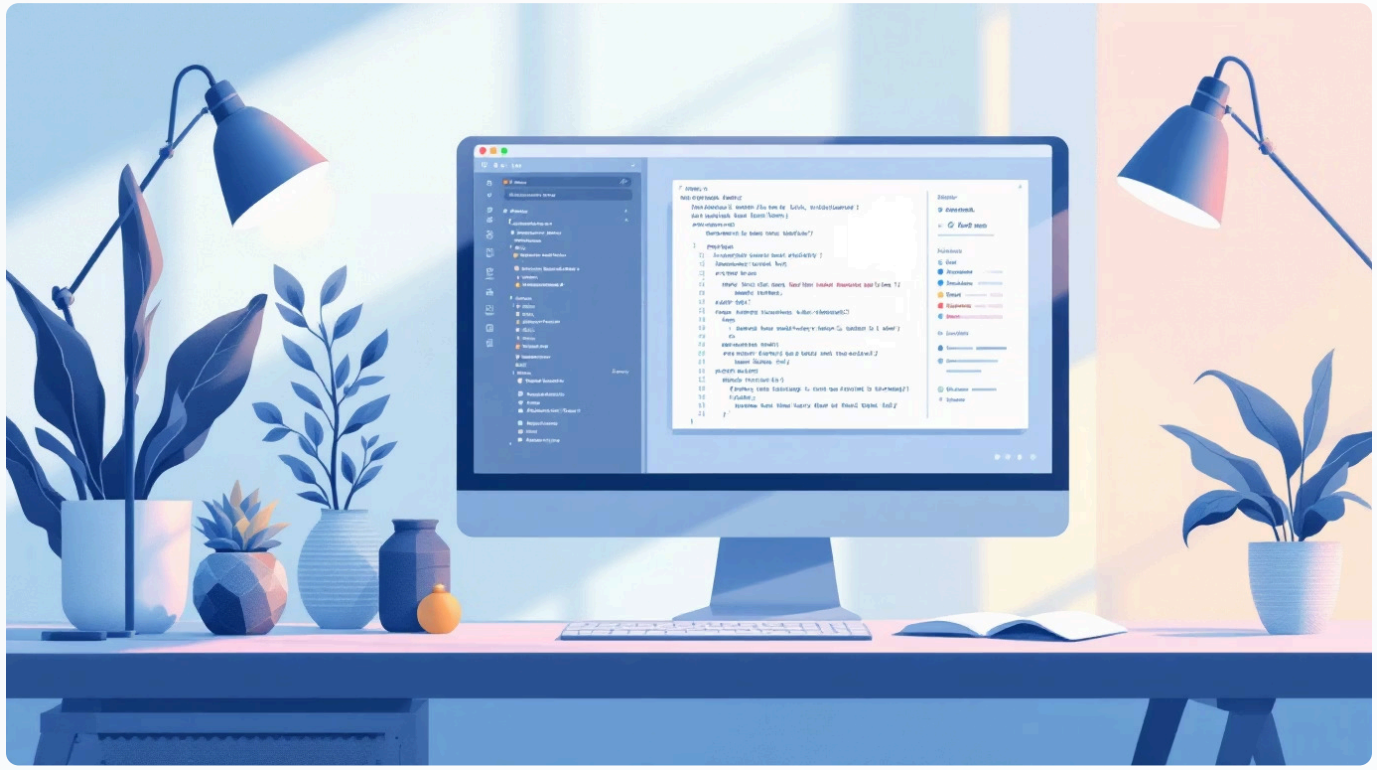
Project Structure & Implementation

File Structure

- data/ folder
- notebooks/ folder
- clustering.py
- README.md

Run Commands

```
git clone https://github.com/your-username/customer-segmentation.git
cd customer-segmentation
pip install pandas numpy matplotlib seaborn scikit-learn
python clustering.py
```



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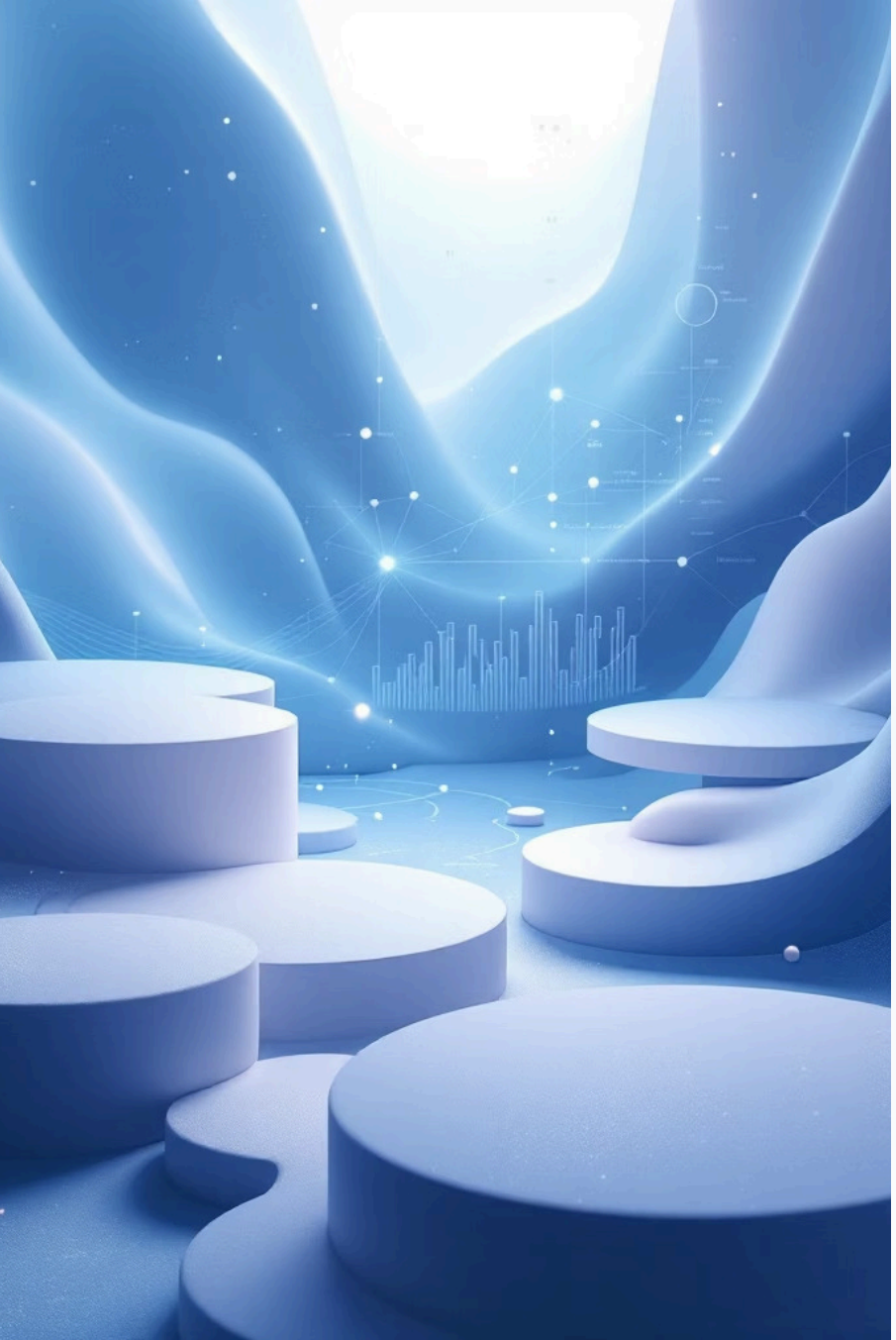

Future Improvements

Real-World Data

Use actual customer datasets

Advanced Clustering

DBSCAN and hierarchical methods

Feature Scaling

Improved accuracy

Interactive App

Deploy with Streamlit

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